

Backup and Disaster Recovery Plan – Ubuntu Innovations

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- Purpose

This document will give you all steps for the backup and disaster recovery process at Ubuntu. The goal here is to make sure the business keeps running in cases of hardware failure, load shedding, data loss, hacking as well as natural disaster.

- What gets backed up

Device	IP Address	Data to Back Up	Backup Frequency	Device
NAS Device	192.168.1.10	All shared folders (/Company/Public, /Finance, /HR, /Sales, /Development, /Management)	Daily incremental, Weekly full	NAS Device
Local Server	192.168.1.11	Databases (MySQL/PostgreSQL), GitLab repositories, Docker volumes	Daily incremental	Local Server
Desktop PCs (20)	192.168.1.50-200	User profiles (Documents, Desktop, Downloads)	Weekly (user-initiated optional)	Desktop PCs (20)
Laptops (5)	192.168.1.50-200	User profiles (Documents, Desktop, Downloads)	Weekly (user-initiated optional)	Laptops (5)

- Backup Schedule

Backup Type	Frequency	Time	Storage Location	Retention Period
Incremental	Daily (Monday to Friday)	2:00 AM	Local NAS (RAID 1)	30 days
Full	Weekly	Sunday at 1:00 AM	Local NAS + Cloud	4 weeks (local), 12 months (cloud)
Monthly Archive	Monthly	1st Sunday of month at 2:00 AM	Cloud only (AWS S3 / Backblaze)	12 months

- Storage Locations

Location	Device	Capacity	Purpose
Primary (Local)	Synology NAS (8TB RAID 1)	4TB usable (mirrored)	Daily and weekly backups
Secondary (Local)	Local server (2x1TB SSD RAID 1)	1TB usable	Critical database and GitLab backups only
Tertiary (Cloud)	AWS S3 or Backblaze B2	2TB	Offsite monthly archives and disaster recovery

- Recovery Objectives

Objective	Target	Explanation
RTO (Recovery Time Objective)	4 hours	Maximum time allowed to restore critical systems after a disaster
RPO (Recovery Point Objective)	1 hour	Maximum acceptable data loss (1 hour of work max)

- Recovery Procedures

a. Single file accidentally deleted

Step	Action	Responsible
1	User contacts IT Support	User
2	IT Support opens NAS backup interface (or cloud backup)	IT Support
3	Locate the file from the most recent backup	IT Support

4	Restore file to original location	IT Support
5	Confirm with user that file is recovered	IT Support

b. NAS device fails by crash

Step	Action	Responsible
1	IT Support detects NAS failure via monitoring alert	IT Support
2	Replace failed hard drive (RAID 1 allows hot swap)	IT Support
3	NAS automatically rebuilds data from remaining drive	Automatic
4	If both drives fail, restore from cloud backup	IT Support
5	Verify shared folders are accessible to all departments	IT Support

c. Digital attack (files encrypted)

Step	Action	Responsible
1	Identify infected device(s) and disconnect from network	IT Support
2	Identify the last known clean backup	IT Support

3	Wipe infected devices completely	IT Support
4	Restore all data from clean backup (cloud or NAS)	IT Support
5	Scan all restored files for malware	IT Support
6	Reconnect devices and reset user passwords	IT Support + Users

d. Load shedding

Step	Action	Responsible
1	UPS (1500VA) keeps router, switch, NAS, and local server running	Automatic
2	UPS provides 15-30 minutes of runtime	Automatic
3	If power not restored within 15 minutes, IT Support initiates graceful shutdown of NAS and server	IT Support
4	When power returns, systems auto-restart	Automatic
5	Verify all services are operational	IT Support

e. Fire or flood, lockdown etc disaster

Step	Action	Responsible
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1	Employees work from home (remote)	Management
2	Access cloud backups (AWS S3 / Backblaze) from any location	IT Support
3	Set up temporary VPN access to cloud-hosted critical data	IT Support
4	Restore essential files to employee laptops	IT Support + Users
5	When office is accessible, rebuild local infrastructure and restore from cloud	IT Support

- Backup testing

Test Type	Frequency	Procedure	Responsible
Automated integrity check	Daily	Backup software verifies backup files are readable	Automatic
Test file restore	Weekly	Restore one random file from backup to verify	IT Support
Full disaster recovery drill	Quarterly (Jan, Apr, Jul, Oct)	Simulate NAS failure and restore from cloud	IT Support + Management

- Responsibilities

Role	Responsibilities
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IT Support	Perform daily backup monitoring, weekly test restores, quarterly recovery drills
Development Team Lead	Ensure local server backups are working (databases, GitLab)
All Employees	Save important files to shared folders (NAS), not local desktops